

Problem Statement

The objective of this analysis is to understand Netflix's content distribution across movies and TV shows, identify trends in content release timings, genre, countries, actors, directors and to derive insight from these analysis that can help Netflix decide what content to produce for various regions.

✎ Importing Python libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

✎ Basic analysis of data

```
netflix = pd.read_csv('netflix.csv')
```

```
netflix.shape
```

```
(8807, 12)
```

Dataset Shape: Rows: 8807 Columns: 12

✎ Missing Value detection

```
netflix.isna().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

✎ Datatypes:

1. Categorical: type, director, cast, country, rating, listed_in
2. Numerical: release_year
3. Date: date_added
 - Converted date_added to datetime
 - Converted Categorical columns to category

Missing Value Analysis:

- Replaced missing Nan values with 'Unknown Coulmn_name' for better analysis

Un-nesting of data:

- Expanded coulmnns which were having multiple comma separated values in a single coulumn and exploded it for better analysis.
The columns include cast, director, country, listed_in.

```
netflix['cast'] = netflix['cast'].str.split(',')
netflix['listed_in'] = netflix['listed_in'].str.split(',')
netflix['country'] = netflix['country'].str.split(',')
netflix['director'] = netflix['director'].str.split(',')
netflix = netflix.explode('cast')
netflix = netflix.explode('listed_in')
netflix = netflix.explode('country')
netflix = netflix.explode('director')
netflix['director'] = netflix['director'].fillna('Unknown Director')
netflix['cast'] = netflix['cast'].fillna('Unknown Cast')
netflix['country'] = netflix['country'].fillna('Unknown Country')
netflix['date_added'] = netflix['date_added'].fillna('Unknown Date Added')
netflix['rating'] = netflix['rating'].fillna('Unknown Rating')
netflix['duration'] = netflix['duration'].fillna('Unknown Duration')
netflix['cast'] = netflix['cast'].str.strip()
netflix['listed_in'] = netflix['listed_in'].str.strip()
netflix['country'] = netflix['country'].str.strip()
netflix['director'] = netflix['director'].str.strip()
```

```
netflix.isna().sum()
```

	0
show_id	0
type	0
title	0
director	0
cast	0
country	0
date_added	0
release_year	0
rating	0
duration	0
listed_in	0
description	0

dtype: int64

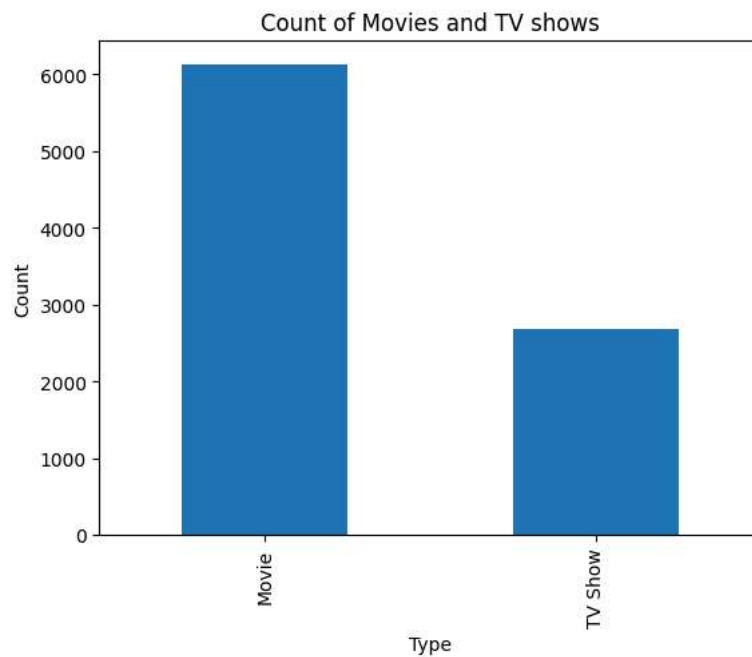
1. Find the counts of each categorical variable both using graphical and nongraphical analysis.
- Content Type
 - a. Non-graphical Analysis:

```
netflix.drop_duplicates('show_id')['type'].value_counts(ascending = False)
```

```
      count
type
Movie    6131
TV Show   2676
dtype: int64
```

✓ b. Graphical analysis:

```
netflix.drop_duplicates('show_id')['type'].value_counts().plot(kind = 'bar')
plt.xlabel('Type')
plt.ylabel('Count')
plt.title('Count of Movies and TV shows')
plt.show()
```



Analysis and Interpretation

From non graphical analysis using value counts and graphical analysis using count plot in type column, it is known that the movies is most preferred by Netflix. The number of movies in Netflix is nearly 3 times more than the TV shows. This indicates that Netflix currently focuses on movie content rather than TV shows

✓ Directors

✓ a. Non-graphical Analysis:

```
netflix['director'].value_counts(ascending = False)
```

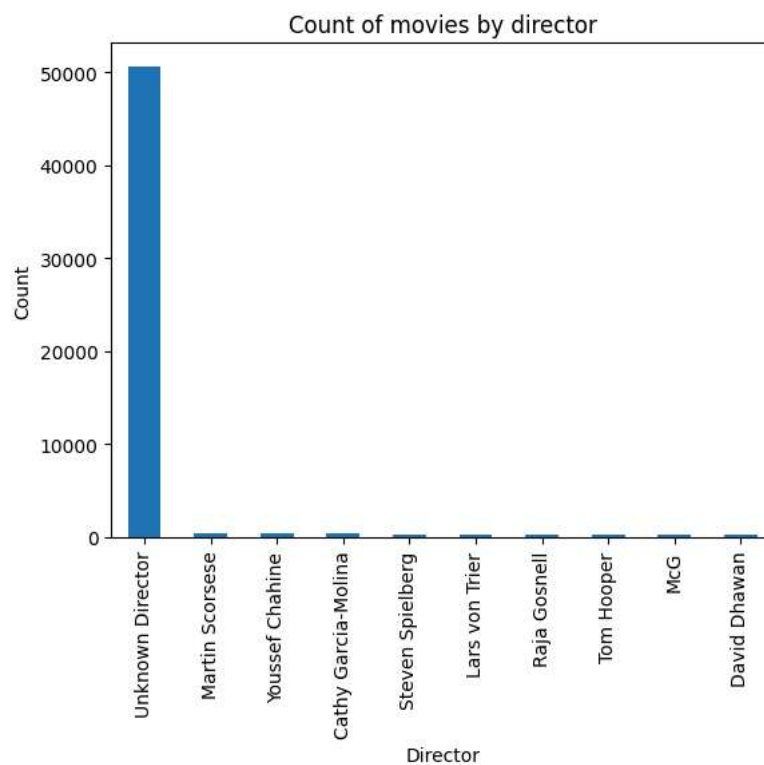
count	
director	
Unknown Director	50643
Martin Scorsese	419
Youssef Chahine	409
Cathy Garcia-Molina	356
Steven Spielberg	355
...	...
Ben Patterson	1
Jedd Wider	1
Todd Wider	1
Alma Har'el	1
Michael Bonfiglio	1

4994 rows × 1 columns

dtype: int64

✓ b. Graphical analysis:

```
netflix['director'].value_counts(ascending = False).head(10).plot(kind = 'bar')
plt.xlabel('Director')
plt.ylabel('Count')
plt.title('Count of movies by director')
plt.show()
```



Analysis and Interpretation

From the non graphical analysis using value counts and graphical analysis using count plots on 'director' column, it is known that the 'Unknown Director' category has the highest number of counts which indicates that the Netflix dataset has significant number of missing values. Excluding this category we can see that the Martin Scorsese has directed highest number of films closely followed by Youssef Chahine.

✓ Cast

a. Non-graphical Analysis:

```
netflix['cast'].value_counts(ascending = False)
```

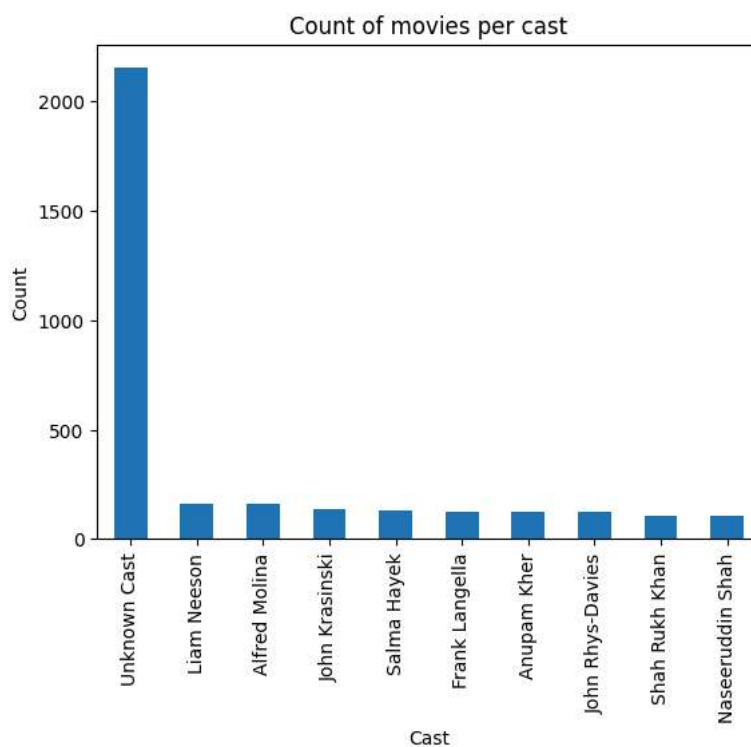
count	
cast	
Unknown Cast	2149
Liam Neeson	161
Alfred Molina	160
John Krasinski	139
Salma Hayek	130
...	...
Kelis	1
Elias Toufexis	1
Claudia Christian	1
Joe Flanigan	1
Bianca Brigitte VanDamme	1

36440 rows × 1 columns

dtype: int64

b. Graphical Analysis:

```
netflix['cast'].value_counts(ascending = False).head(10).plot(kind = 'bar')  
plt.xlabel('Cast')  
plt.ylabel('Count')  
plt.title('Count of movies per cast')  
plt.show()
```



Analysis and Interpretation

From the Non graphical analysis using value counts and graphical analysis using count plot, it can be inferred that 'Unknown Cast' dominates the dataset. This clearly indicates the presence of a significant amount of missing cast information in the netflix dataset.

After excluding this category, Liam Neeson is identified as the actor who has appeared in highest number of movies, followed by Alfred Molina.

Country

a. Non-graphical Analysis:

```
netflix['country'].value_counts(ascending = False)
```

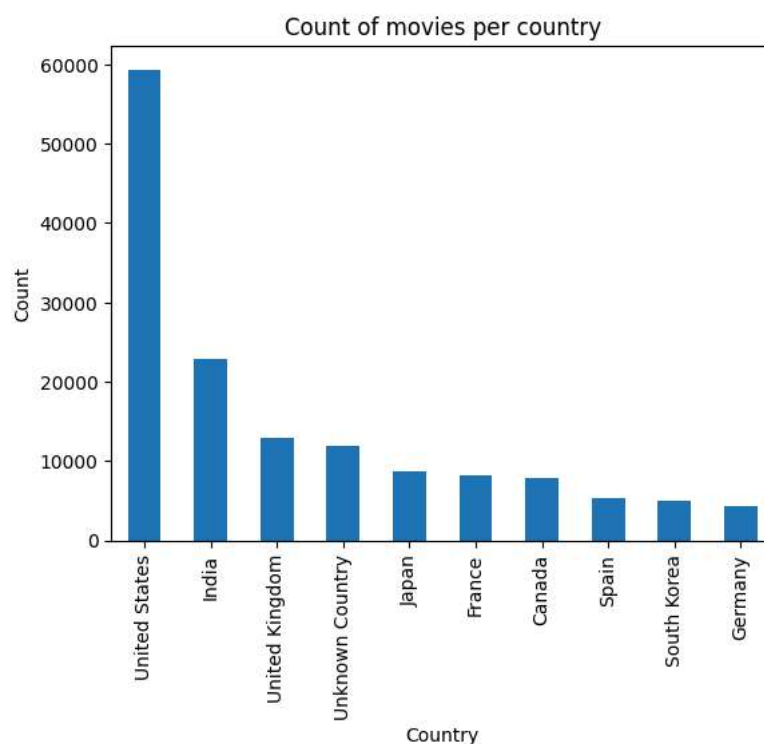
country	count
United States	59350
India	22814
United Kingdom	12965
Unknown Country	11897
Japan	8679
...	...
Botswana	2
Afghanistan	2
Nicaragua	1
Kazakhstan	1
Uganda	1

124 rows × 1 columns

dtype: int64

b. Graphical Analysis:

```
netflix['country'].value_counts(ascending = False).head(10).plot(kind = 'bar')  
plt.xlabel('Country')  
plt.ylabel('Count')  
plt.title('Count of movies per country')  
plt.show()
```



Analysis and Interpretation

From non-graphical analysis using value counts and graphical analysis using count plot, it is observed that the United States has contributed the highest number of movies/tv shows to the netflix dataset. The count of movies/tv shows of the United States is nearly 3 times higher than the second ranked country, India.

▼ Date Added

▼ a. Non-graphical Analysis:

```
netflix.drop_duplicates('show_id')['date_added'].value_counts(ascending = False)
```

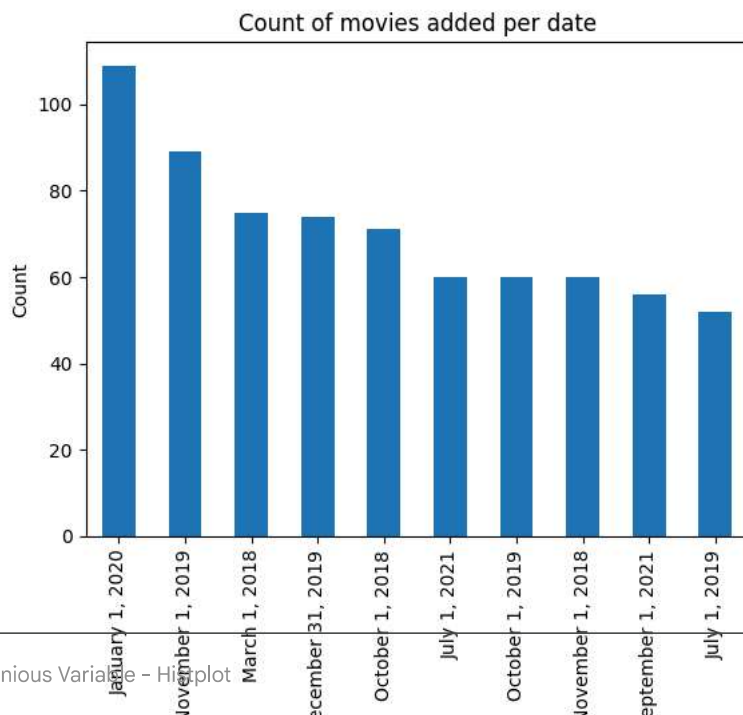
date_added	count
January 1, 2020	109
November 1, 2019	89
March 1, 2018	75
December 31, 2019	74
October 1, 2018	71
...	...
June 1, 2018	1
October 21, 2018	1
February 8, 2016	1
June 5, 2018	1
October 14, 2017	1

1768 rows × 1 columns

dtype: int64

▼ b. Graphical Analysis:

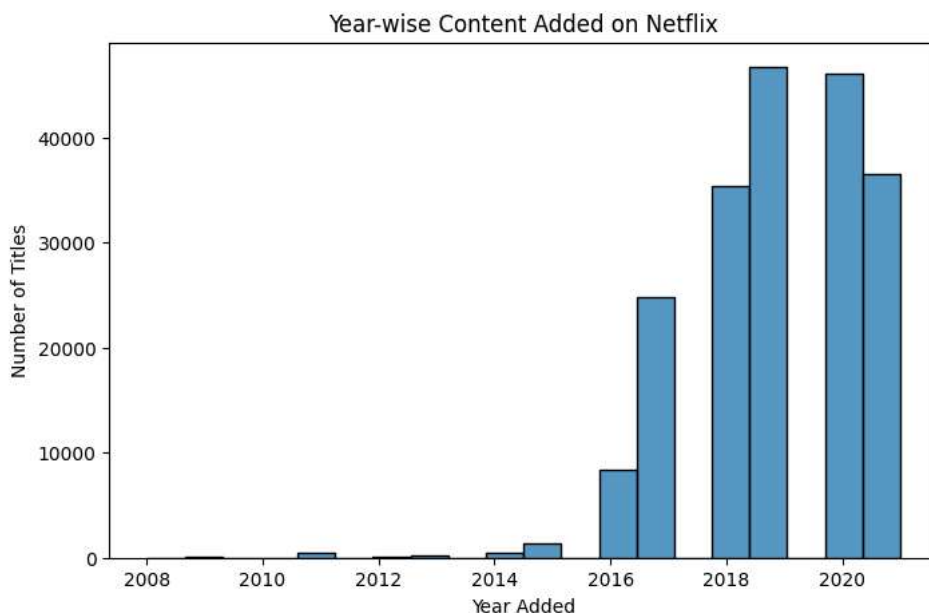
```
netflix.drop_duplicates('show_id')['date_added'].value_counts(ascending = False).head(1)
plt.xlabel('Date Added')
plt.ylabel('Count')
plt.title('Count of movies added per date')
plt.show()
```



Continuous Variable - Histogram

```
netflix['date_added'] = pd.to_datetime(netflix['date_added'], errors= 'coerce')
netflix['added_year'] = netflix['date_added'].dt.year
```

```
plt.figure(figsize=(8,5))
sns.histplot(netflix['added_year'].dropna(), bins=20)
plt.title('Year-wise Content Added on Netflix')
plt.xlabel('Year Added')
plt.ylabel('Number of Titles')
plt.show()
```



Analysis and Interpretation

From non graphical analysis using value counts and graphical analysis using count plot, it is observed that January 01, 2020 recorded highest number of movies and tv shows added to the Netflix platform. This suggests that the netflix tends to add large volume at the beginning of the year.

The Histogram distribution shows sharp increase in the number of titles added after 2016 indicating the Netflix's rapid expansion phase. This shows the platform's aggressive growth in recent years.

Release Year

▼ a. Non-graphical Analysis:

```
netflix.drop_duplicates('show_id')['release_year'].value_counts(ascending = False)
```

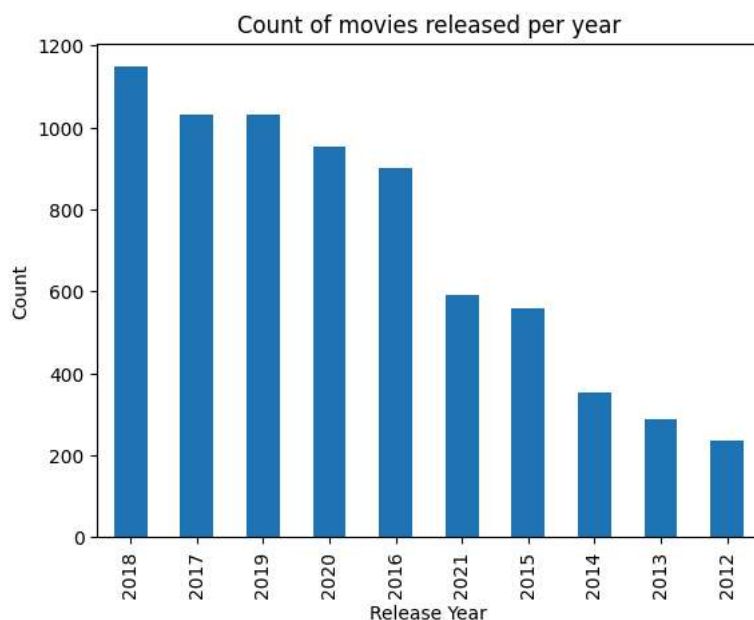
release_year	count
2018	1147
2017	1032
2019	1030
2020	953
2016	902
...	...
1961	1
1925	1
1959	1
1966	1
1947	1

74 rows × 1 columns

dtype: int64

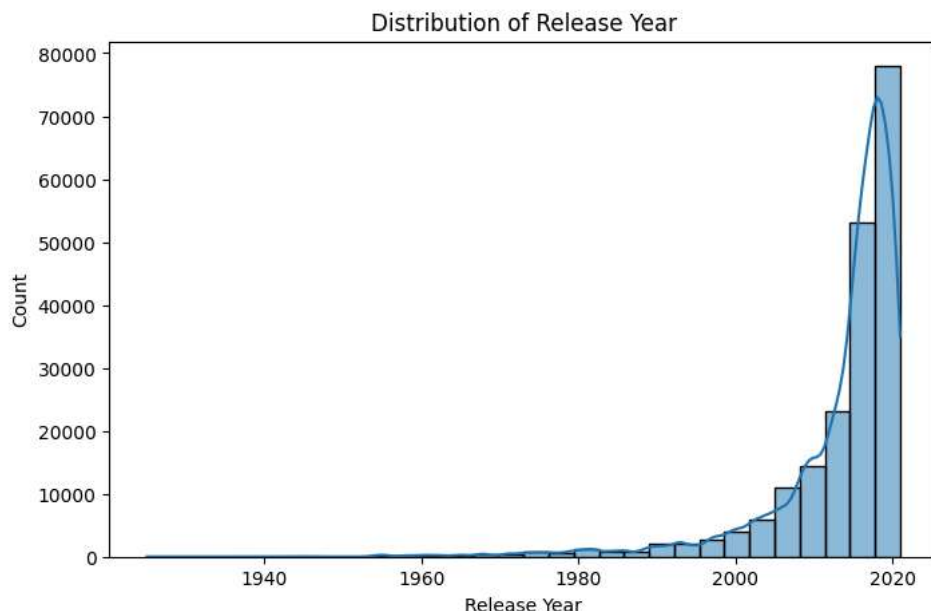
▼ b. Graphical Analysis:

```
netflix.drop_duplicates('show_id')['release_year'].value_counts(ascending = False).head(10)
plt.xlabel('Release Year')
plt.ylabel('Count')
plt.title('Count of movies released per year')
plt.show()
```



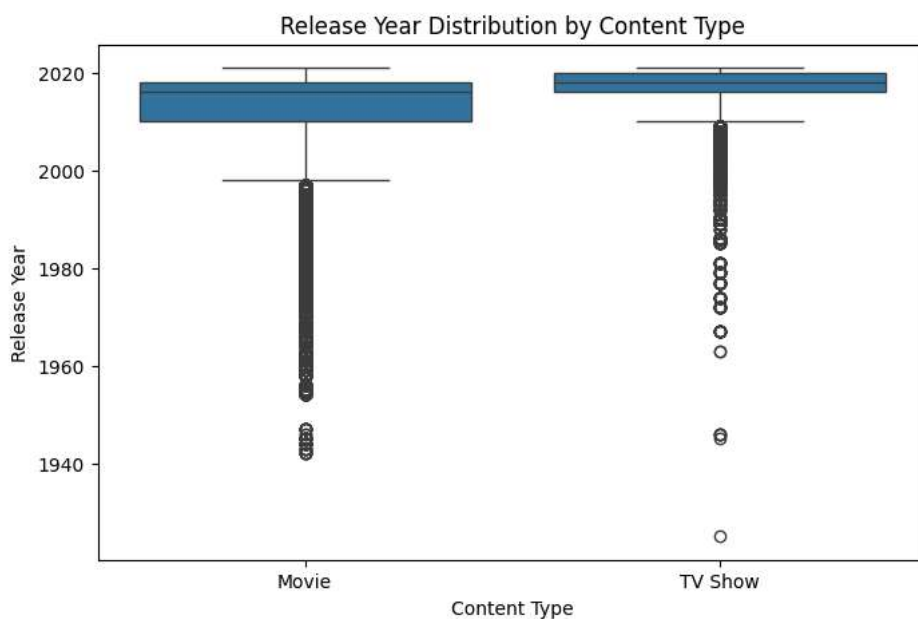
Continuous Variable - Histplot

```
plt.figure(figsize=(8,5))
sns.histplot(netflix['release_year'], bins=30, kde=True)
plt.title('Distribution of Release Year')
plt.xlabel('Release Year')
plt.ylabel('Count')
plt.show()
```



Categorical Variables - BoxPlot

```
plt.figure(figsize=(8,5))
sns.boxplot(x='type', y='release_year', data=netflix)
plt.title('Release Year Distribution by Content Type')
plt.xlabel('Content Type')
plt.ylabel('Release Year')
plt.show()
```



Analysis and Interpretation

From non graphical analysis using value counts and non graphical analysis using count plots, it is observed that the year 2018 has recorded the highest number of movie and tv show releases on Netflix. The Histplot shows a right skewed distribution of the release years with significant increase in the content after the year 2000. The peak in recent years indicate the Netflix's focus on accruing new content.

The Boxplot shows that the TV shows have higher median release year when compared to movies indicating the TV shows are more recent. Movies show wider spread which indicates mix of both old and new content. The outliers indicate the presence of few very old titles.

Rating

v a. Non-graphical Analysis:

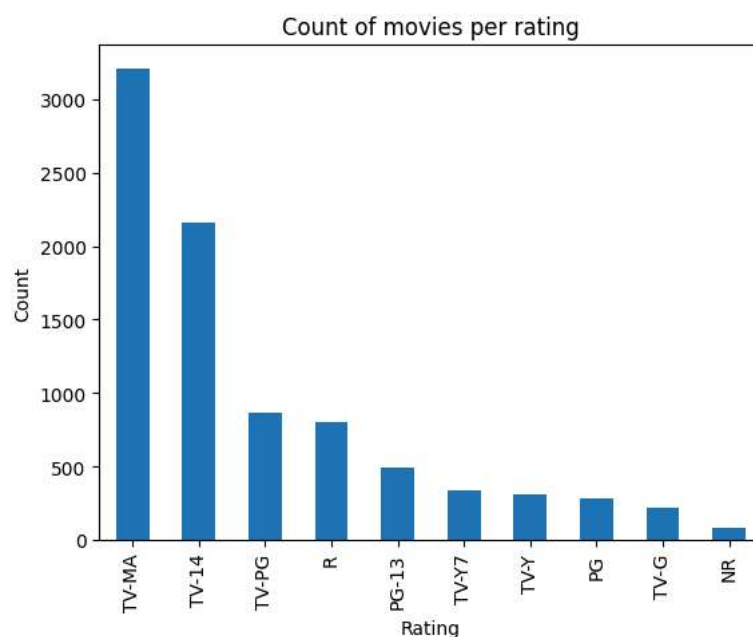
```
netflix.drop_duplicates('show_id')['rating'].value_counts(ascending = False)
```

rating	count
TV-MA	3207
TV-14	2160
TV-PG	863
R	799
PG-13	490
TV-Y7	334
TV-Y	307
PG	287
TV-G	220
NR	80
G	41
TV-Y7-FV	6
Unknown Rating	4
NC-17	3
UR	3
66 min	1
74 min	1
84 min	1

dtype: int64

v b. Graphical Analysis:

```
netflix.drop_duplicates('show_id')['rating'].value_counts(ascending = False).head(10).  
plt.xlabel('Rating')  
plt.ylabel('Count')  
plt.title('Count of movies per rating')  
plt.show()
```



Analysis and Interpretation

From non graphical analysis using value counts and graphical analysis using count plots, it is observed that the TV-MA rating has the highest count of movies and tv shows in the Netflix. This indicates that the Netflix prefers the content targeted for mature audiences when compared to other rating categories.

Duration

a. Non-graphical Analysis:

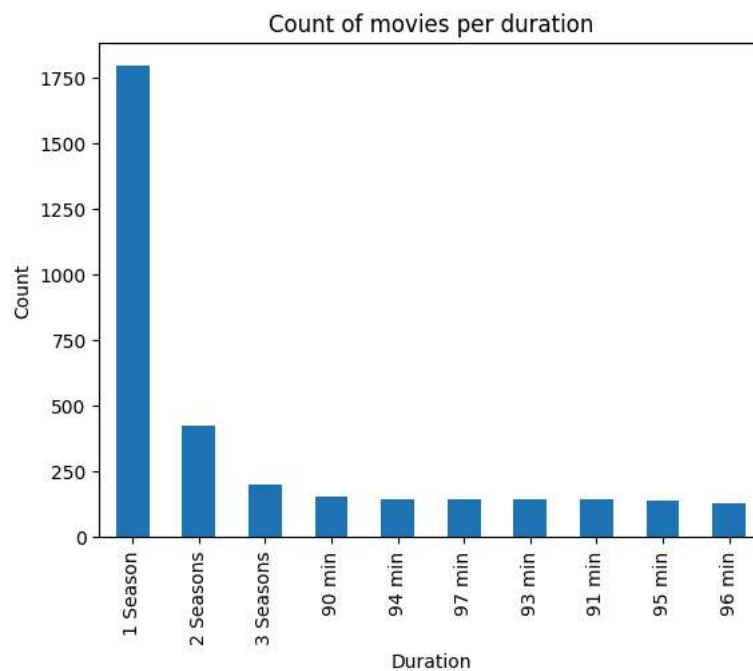
```
netflix.drop_duplicates('show_id')['duration'].value_counts(ascending = False)
```

	count
1 Season	1793
2 Seasons	425
3 Seasons	199
90 min	152
94 min	146
...	...
228 min	1
18 min	1
205 min	1
201 min	1
191 min	1

221 rows × 1 columns

dtype: int64

```
netflix.drop_duplicates('show_id')['duration'].value_counts(ascending = False).head(10)
plt.xlabel('Duration')
plt.ylabel('Count')
plt.title('Count of movies per duration')
plt.show()
```

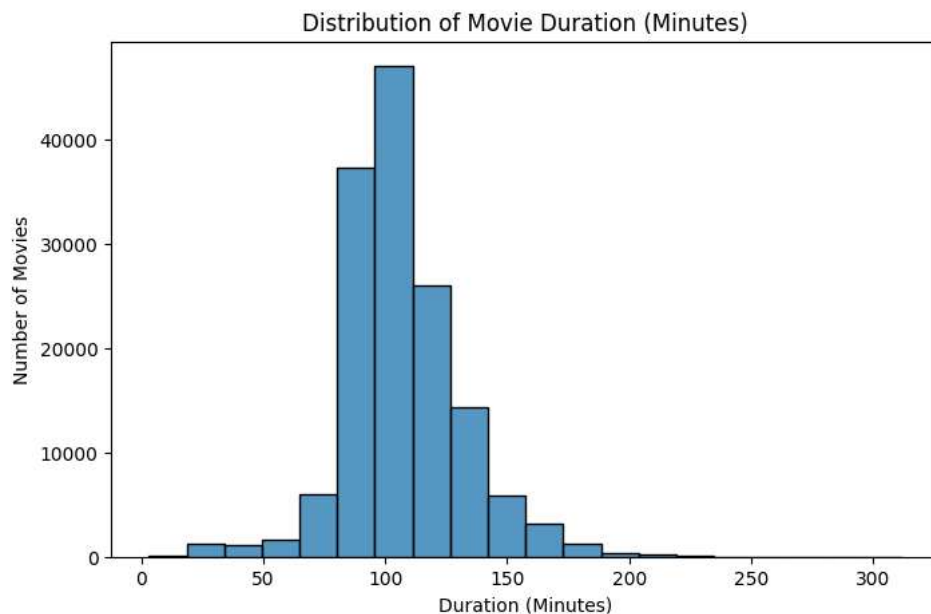


Histplot

```

movies = netflix[(netflix['type'] == 'Movie') & (netflix['duration'].str.contains('min'))
movies['duration_min'] = movies['duration'].str.replace(' min', '', regex=False).astype(int)
plt.figure(figsize=(8,5))
sns.histplot(movies['duration_min'], bins=20)
plt.title('Distribution of Movie Duration (Minutes)')
plt.xlabel('Duration (Minutes)')
plt.ylabel('Number of Movies')
plt.show()

```



Analysis and Interpretation

From non graphical using value counts and non graphical analysis using count plots, it is observed that the "1 season" is the most common duration for the tv shows in Netflix. This indicates that Netflix has large number of TV shows with single season which may be preferred by those who like consise content.

The Histogram shows that most movies in netflix has the duration between 80 and 120 mins. This indicates the preference for regular standard length movie.

✓ Listed in

✓ a. Non-graphical Analysis:

```
netflix['listed_in'].value_counts(ascending = False)
```

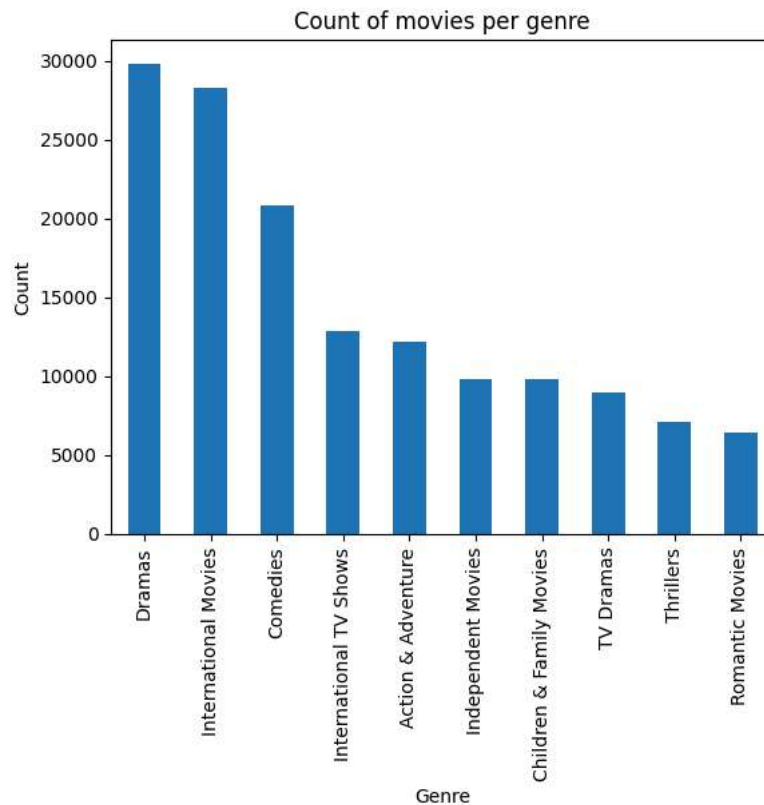
	count
listed_in	
Dramas	29806
International Movies	28243
Comedies	20829
International TV Shows	12845
Action & Adventure	12216
Independent Movies	9834
Children & Family Movies	9771
TV Dramas	8942
Thrillers	7107
Romantic Movies	6412
TV Comedies	4963
Crime TV Shows	4733
Horror Movies	4571
Kids' TV	4568
Sci-Fi & Fantasy	4037
Music & Musicals	3077
Romantic TV Shows	3049
Documentaries	2409
Anime Series	2313
TV Action & Adventure	2288
Spanish-Language TV Shows	2126
British TV Shows	1808
Sports Movies	1531
Classic Movies	1443
TV Mysteries	1281
Korean TV Shows	1122
Cult Movies	1077
Anime Features	1045
TV Sci-Fi & Fantasy	1045
TV Horror	941
Docuseries	845
LGBTQ Movies	838
TV Thrillers	768
Teen TV Shows	742
Reality TV	735
Faith & Spirituality	719
Stand-Up Comedy	540
Movies	412
TV Shows	337
Classic & Cult TV	272
Stand-Up Comedy & Talk Shows	268
Science & Nature TV	157

dtype: int64

∨ b. Graphical analysis:

```
netflix['listed_in'].value_counts(ascending = False).head(10).plot(kind = 'bar')
plt.xlabel('Genre')
```

```
plt.ylabel('Count')
plt.title('Count of movies per genre')
plt.show()
```



Analysis and Interpretation

From non graphical analysis using value counts and graphical analysis using count plots, it is observed that the highest count of movies and TV shows are in the genre "Dramas". This indicates that Netflix has strong emphasis of drama oriented content compared to other genre.

2. Comparison of tv shows vs. movies.

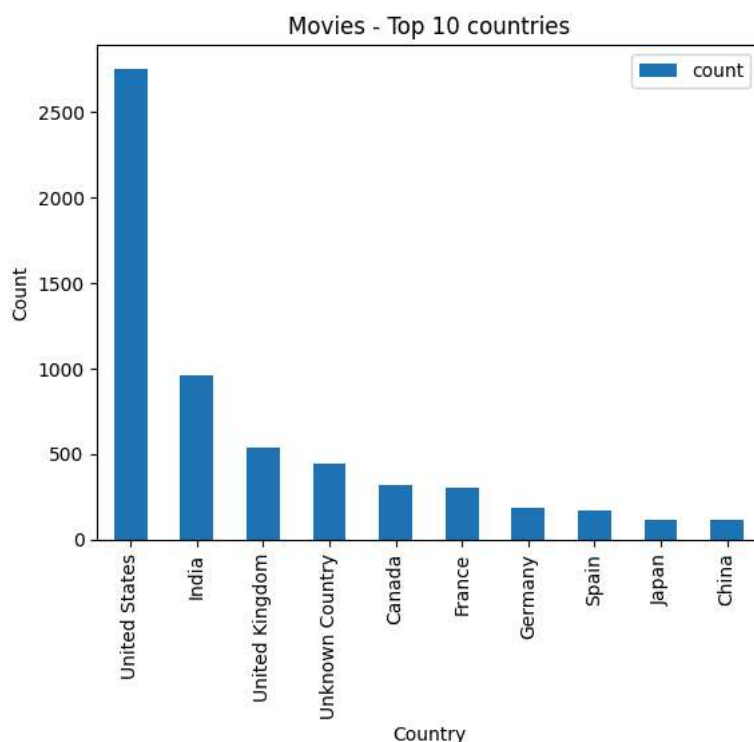
- a. Find the number of movies produced in each country and pick the top 10 countries.

```
top_10_countries = netflix[netflix['type']=='Movie'].groupby('country')['title'].nunique()
top_10_countries
```

	country	count
0	United States	2752
1	India	962
2	United Kingdom	534
3	Unknown Country	440
4	Canada	319
5	France	303
6	Germany	182
7	Spain	171
8	Japan	119
9	China	114

Next steps: [Generate code with top_10_countries](#) [New interactive sheet](#)

```
top_10_countries.plot(kind = 'bar', x= 'country', y = 'count')
plt.xlabel('Country')
plt.ylabel('Count')
plt.title('Movies - Top 10 countries')
plt.show()
```



Analysis and Interpretation

From non graphical analysis using groupby, nunique and graphical analysis using count plots, it is observed that the country "United States" has dominated in terms of movies produced and available on Netflix. The list of top 10 countries contributing the highest number of movies are given above.

- ✓ b. Find the number of Tv-Shows produced in each country and pick the top 10 countries.

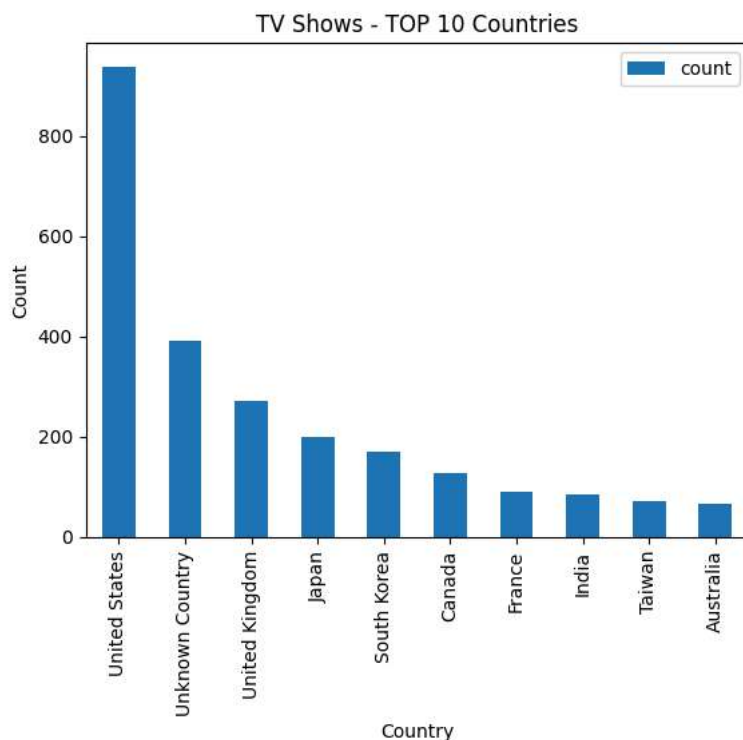
```
top_10_tvshows = netflix[netflix['type']== 'TV Show'].groupby('country')['title'].nunique()
top_10_tvshows
```

	country	count
0	United States	938
1	Unknown Country	391
2	United Kingdom	272
3	Japan	199
4	South Korea	170
5	Canada	126
6	France	90
7	India	84
8	Taiwan	70
9	Australia	66

Next steps: [Generate code with top_10_tvshows](#) [New interactive sheet](#)

```
top_10_tvshows.plot(kind = 'bar', x = 'country', y = 'count')
plt.xlabel('Country')
plt.ylabel('Count')
plt.title('TV Shows - TOP 10 Countries')
```


Text(0.5, 1.0, 'TV Shows - TOP 10 Countries')



Analysis and Interpretation

From non graphical analysis using groupby, nunique and graphical analysis using count plots, it is observed that the country "United States" has dominated in terms of TV shows produced and available on Netflix. The list of top 10 countries contributing the highest number of TV shows are given above.

3. What is the best time to launch a TV show/Movie?

a. Find which is the best week to release the TV shows/movie.

```
netflix['date_added'] = pd.to_datetime(netflix['date_added'], errors= 'coerce')
```

```
netflix['week'] = netflix['date_added'].dt.isocalendar().week
```

Best week to launch a Movie

```
best_week_movie = netflix[netflix['type']=='Movie'].groupby('week')['title'].nunique()
best_week_movie.reset_index()
```

week	title
0	1
1	316

Analysis and Interpretation

From non graphical analysis using groupby, nunique, it is observed that the week 1 has the highest number of titles added to Netflix platform. This indicates that week 1 is the most common and optimal week to add the movie in Netflix.

Best week to launch a TV Shows

```
best_week_tvshow = netflix[netflix['type']=='TV Show'].groupby('week')['title'].nunique()
best_week_tvshow.reset_index()
```

	week	title
0	27	85

Analysis and Interpretation

From non graphical analysis using groupby, nunique, it is observed that the week 27 has the highest number of titles added to Netflix platform. This indicates that week 1 is the most common and optimal week to add the TV shows in Netflix.

✓ b. Find which is the best month to release the Tv-show/Movies

```
netflix['month'] = netflix['date_added'].dt.month_name()
```

✓ Best month to launch a Movie

```
best_month_movie = netflix[netflix['type']=='Movie'].groupby('month')['title'].nunique()
best_month_movie.reset_index()
```

	month	title
0	July	565

Analysis and Interpretation

From non graphical analysis using groupby, nunique, it is observed that the month "July" has the highest number of titles added to Netflix platform. This indicates that the month "July" is the most common and optimal month to add the movie in Netflix.

✓ Best month to launch TV Show

```
best_month_tvshow = netflix[netflix['type']=='TV Show'].groupby('month')['title'].nunique()
best_month_tvshow.reset_index()
```

	month	title
0	July	254

Analysis and Interpretation

From non graphical analysis using groupby, nunique, it is observed that the month "July" has the highest number of titles added to Netflix platform. This indicates that the month "July" is the most common and optimal month to add the TV Shows in Netflix.

✓ 4. Analysis of actors/directors of different types of shows/movies.

✓ a. Identify the top 10 actors who have appeared in most movies or TV shows.

```
top_10_actors = netflix.groupby('cast')['title'].nunique().sort_values(ascending = False)
top_10_actors.reset_index().rename(columns = {'title':'count'})
```

	cast	count	
0	Unknown Cast	825	
1	Anupam Kher	43	
2	Shah Rukh Khan	35	
3	Julie Teiwani	33	
4	Naseeruddin Shah	32	
5	Takahiro Sakurai	32	
6	Rupa Bhimani	31	
7	Akshay Kumar	30	
8	Om Puri	30	
9	Yuki Kaji	29	

Analysis and Interpretation

From non graphical analysis using groupby and nunique, it is observed that 'Unknown Cast' dominates the Netflix dataset. This indicates that there is significant number of missing data of cast information. Excluding this category, Anupam Kher has appeared in highest number of movies and TV shows. The list of top 10 actors is mentioned above.

- ✓ b. Identify the top 10 directors who have appeared in most movies or TV shows

```
top_10_directors = netflix.groupby('director')['title'].nunique().sort_values(ascending=False)
top_10_directors.reset_index()
```

	director	title	
0	Unknown Director	2634	
1	Rajiv Chilaka	22	
2	Jan Suter	21	
3	Raúl Campos	19	
4	Marcus Raboy	16	
5	Suhas Kadav	16	
6	Jay Karas	15	
7	Cathy Garcia-Molina	13	
8	Youssef Chahine	12	
9	Jay Chapman	12	

Analysis and Interpretation

From non graphical analysis using groupby and nunique, it is observed that 'Unknown Director' dominates the Netflix dataset. This indicates that there is significant number of missing data of director information. Excluding this category, Rajiv Chilaka has directed highest number of movies and TV shows. The list of top 10 directors is mentioned above.

- ✓ 5. Which genre movies are more popular or produced more

```
from wordcloud import WordCloud
text = netflix['listed_in'].value_counts()
```

```
wordcloud = WordCloud(width= 800, height= 400, background_color= 'white', stopwords= {
plt.figure(figsize= (12,6))
plt.imshow(wordcloud, interpolation= 'bilinear')
plt.axis('off')
plt.title('Word cloud of Netflix Genres')
plt.show()
```

Analysis and Interpretation

From graphical analysis using wordcloud, it can be observed that the genre "Dramas" appears in the largest font size in the word cloud. This indicates that the genre has highest frequency of occurrence in the Netflix dataset. This shows that there is a strong and consistent audience for drama based content

6. Find After how many days the movie will be added to Netflix after the release of the movie (you can consider the recent past data)

```
netflix['release_year'] = pd.to_datetime(netflix['release_year'].astype('str'), format='%Y')
movies_df = netflix[netflix['type']=='Movie'].drop_duplicates('show_id')
recent_movies = movies_df[movies_df['release_year']>= '2018-01-01'].copy()
recent_movies['date_to_add'] = (recent_movies['date_added'] - recent_movies['release_year']).dt.days
recent_movies = recent_movies[recent_movies['date_to_add']>0]
best time to add = recent_movies['date to add'].mode()[0]
```

```
print(best time to add)
```

334

Analysis and Interpretation

From non graphical analysis using mode of time difference between date added and release year, it is observed that the optimal time to add new content is after 334 days after the release date. Since the dataset has release date has only release year , default date of January 01 of the year ("YEAR-01-01") was assumed for analysis. The mode of the difference between release date and date added was applied.

Business Insights

- ## 1. Movies Dominate the Netflix content library

The movies are nearly 3 times more than the TV shows which clearly shows that the Netflix focuses more of movies than TV shows.

2. Dramas are most popular genre on Netflix.

Based on the analysis using word cloud, the Dramas are most preferred genre on Netflix which suggest that there is strong audience for drama based content.

3. The United states is the leading content producer.

The United States produces highest number of movies and TV shows available on Netflix. This is followed by India. This highlights the significance of these markets for Netflix.

4. Netflix content addition has increased significantly in recent years

The content addition by Netflix has raised significantly after 2016, indicating Netflix aggressive expansion and investment

5. TV shows are more recent

The Boxplot analysis shows that the TV shows are more recent than movies suggesting that the Netflix is investing in newer TV show content.

6. Most movies have standard duration

The histogram analysis shows that the duration for most movies are between 80 and 120 mins which represent the standard movie duration.

7. January and July are key period for releasing new content

Analysis of week and month of content addition indicate that the week 1 and July month has highest number of title additions which indicates the preferred releasing window.

8. Movies addition on Netflix within a year

The mode of difference between release year and date added is approximately 334 days which indicate that the movies are added within a year of release.

~ Recommendations

Based on the above Business insights, the following actionable recommendations are suggested:

1. Increase investment in drama content

Since the drama content consistently show high presence on Netflix, Netflix should produce and continue to acquire more drama based content on Netflix to meet audience demand.

2. Strength content production in key markets

The Netflix should increase content production in countries such as United States and India since these countries contribute significantly to the platform.

3. Produce newer TV show content

Since the there is more trend for newer TV shows, Netflix should invest in more newer tv shows with limited series content

4. Plan major release during peak month

Netflix should plan major content release during the month of January and July for increasing the customer engagement.

5. Improve metadata quality

Netflix should improve the metadata quality by reducing the missing informations of cast, directors, date added and ratings to increase the user experience.

6. Focusing on Standard length movies

Netflix should focus on producing the movies of standard length between 80 and 120 mins as preferred by the viewers.

7. Invest in TV-MA content

Since there is a huge audience for TV-MA rated content, Netflix should focus on producing more TV-MA rated content

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